NARAYANA IIT ACADEMY INDIA

40+ YEARS OF EXCELLENCE

Sec: OSR.IIT_*CO-SC	GTA-5(P1)	Date: 14-04-24
Time: 3HRS	2016_P1	Max. Marks: 198

KEY SHEET

MATHEMATICS

1	**D**	2	**B**	3	**D**	4	**C**	5	**B**
6	**ABD**	7	**BC**	8	**ABC**	9	**AC**	10	**AD**
11	**ABD**	12	**BC**	13	**C**	14	**5**	15	**2**
16	**1**	17	**0**	18	**8**				

PHYSICS

19	**C**	20	**B**	21	**D**	22	**C**	23	**A**
24	**BC**	25	**ACD**	26	**C**	27	**ABCD**	28	**AC**
29	**AB**	30	**A**	31	**CD**	32	**5**	33	**7**
34	**8**	35	**1**	36	**4**				

CHEMISTRY

37	**C**	38	**C**	39	**B**	40	**B**	41	**B**
42	**ABCD**	43	**BC**	44	**BC**	45	**ABCD**	46	**AD**
47	**BC**	48	**BCD**	49	**BC**	50	**4**	51	**6**
52	**6**	53	**8**	54	**4**				

SOLUTIONS
MATHS

1. $I.F = e^{-\int \frac{x^4+3x^2}{(x^2+1)^2}dx} = e^{\frac{-x^3}{x^2+1}}$

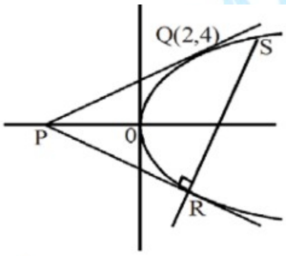
$\Rightarrow ye^{\frac{-x^3}{x^2+1}} = \int (4x+3)dx$

$\Rightarrow f(x) = y = (2x^2+3x+1)e^{\frac{x^3}{x^2+1}}$

$\Rightarrow f(1) = 6\sqrt{e}\ \&\ f(-1) = 0$

$\Rightarrow f(-1) = 0$

2.



$y^2 = 8x, Q(2,4) a = 2$

$\Rightarrow$ parameter of Q is $t = 1$

$\Rightarrow$ parameter of R is $t = -1$

$\Rightarrow$ parameter of S is $t_2 = -t_1 - \frac{2}{t_1}$

i.e. $t_2 = 3$

3.

$$\begin{vmatrix} 2023+2024x^2 & 2023x^2+2024 & 2025 \\ 2017+2018x^2 & 2017x^2+2018 & 2019 \\ 2006+2007x^2 & 2006x^2+2007 & 2008 \end{vmatrix} = 0 \forall x \in R$$

4.

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

Given, Mean and variance are 9 and 15.08 respectively

Mean $= \dfrac{8+16+120+80+56+80+6\alpha+12\beta}{40+\alpha+\beta} = 9$

$\Rightarrow 360+9\alpha+9\beta = 360+6\alpha+12\beta \Rightarrow 3\alpha - 3\beta = 0 \qquad \Rightarrow \alpha = \beta$ .... (1)

Now variance is given by,

$$15.08=\frac{16+64+36\alpha+960+800+144\alpha+784+1280}{40+2\alpha}-9^2$$

$$\Rightarrow 96.08=\frac{3904+180\alpha}{40+2\alpha}\Rightarrow(40+2\alpha)(96.08)=3904+180\alpha$$

$$\Rightarrow 384.3+(192.16)\alpha=3904+180\alpha$$

$$\Rightarrow(12.16)\alpha=60.80 \qquad \Rightarrow\alpha=5=\beta$$

Hence, $\alpha^2+\beta^2+\alpha\beta=25+25+25=75$

5.

$$\text{Coefficient of } x^2=\frac{1}{2}\left[\left(1+2+2^2......+2^n\right)^2\right]-\left[1^2+2^2+....\left(2^n\right)^2\right]$$

$$=\frac{1}{2}\left[\left(\frac{2^{n+1}-1}{1}\right)^2-\left(\frac{\left(2^2\right)^{n+1}-1}{3}\right)\right]=\frac{1}{2}\left[\left(2^{n+1}-1\right)\left[\left(2^{n+1}-1\right)-\frac{2^{n+1}+1}{3}\right]\right]$$

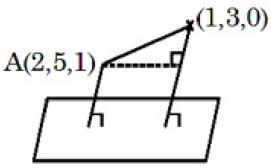
$$=\frac{\left(2^{n+1}-1\right)\left(2^{n+1}-2\right)}{3}$$

6. (A) $\bar{n}_1\times\bar{n}_2=<-2,3,-1>$

Point common to P_1 and P_2 is (1, 2, 0)

$\therefore$ distance between skew lines is

$$\left|\frac{(\bar{a}-\bar{c}).(\bar{p}\times\bar{q})}{|\bar{p}\times\bar{q}|}\right|=\frac{1}{5\sqrt{3}}$$



(B)

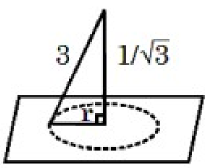
$\overline{BA}=\hat{i}+2\hat{j}+\hat{k}$

Angle between line and plane

$$=\sin^{-1}\left(\frac{4}{\sqrt{3}\sqrt{6}}\right)=\theta$$

Projection is $|\overline{BA}|\cos\theta=\frac{\sqrt{6}\times\sqrt{2}}{\sqrt{18}}=\frac{\sqrt{6}}{3}$

(D) $r=\sqrt{\frac{26}{3}}$



Area $=r=\frac{26\pi}{3}$

7.

$$\sum_{i=0}^{100}\sum_{j=0}^{100}\left(C_i^2+C_j^2+C_iC_j\right)=a\,{}^{2n}C_n+2^b$$

$$\sum_{i=0}^{100}\left(101\,C_i^2+{}^{200}C_{100}+C_i2^{100}\right)$$

$$101\,{}^{200}C_{100}+101\,{}^{200}C_{100}+2^{100}.2^{100}$$

$$202\,{}^{200}C_{100}+2^{200}=a\,{}^{2n}C_n+2^b$$

$a=202$; $n=100$; $b=200$

8. Hence none of z_i are zero and no two of z_i are equal i=1,2,3,4

Let $f(x)=(x-z_1)(x-z_2)(x-z_3)(x-z_4)$

$$f^{|}(x)=(x-z_2)(x-z_3)(x-z_4)+(x-z_1)(x-z_3)(x-z_4)$$
$$+(x-z_1)(x-z_2)(x-z_4)+(x-z_1)(x-z_2)(x-z_3)$$

$$f^{|}(z_1)=(z_1-z_2)(z_1-z_3)(z_1-z_4)=\frac{4}{z_1}$$

Similarly, $f^{|}(z_2)=\frac{4}{z_2}$

$f^{|}(z_3)=\frac{4}{z_3},f^{|}(z_4)=\frac{4}{z_4}$

$\Rightarrow xf^{|}(x)=4$ has roots $z_1z_2z_3$ and z_4

$\Rightarrow xf^{|}(x)-4=0$ is a 4 degree polynomial has leading coefficient is 4

$\Rightarrow xf^{|}(x)-4=4f(x)$

$$\Rightarrow\frac{f^{|}(x)}{f(x)+1}=\frac{4}{x}$$

log(f(x) + 1) = 4logx + a, a be any constant

$\log\left(f(x)+1\right)=\log\left(ax^4\right)$

$f(x)+1=ax^4$

$f(x)=ax^4-1$ comparing coefficient of x^4 we get a = 1

$f(x)=x^4-1$

then the roots are 1, –1, i, –i

9. P(Prime number) $=\frac{1}{2}$

$\therefore p=\frac{1}{2}; q=\frac{1}{2}; n=6$

$P(E)=P(r\geq 3)$

$=\frac{{}^6C_3+{}^6C_4+{}^6C_5+{}^6C_6}{64}$

$=\frac{42}{64}=\frac{21}{32}$

$P(F)=\frac{1}{2}.\frac{1}{2}.\frac{1}{2}=\frac{1}{8}$

$P(F/E)=\frac{P(F\cap E)}{P(E)}=\frac{P(F)}{P(E)}$

$=\frac{1}{8}\left(\frac{32}{21}\right)=\frac{4}{21}=\frac{a}{b}$

$\therefore a=4$ and $b=21$

10. $f(x)=\tan^{-1}\left(\frac{\frac{x}{3}-\frac{x}{\sqrt{3}}}{1+\frac{x^2}{3\sqrt{3}}}\right)=\tan^{-1}\frac{x}{3}-\tan^{-1}\frac{x}{\sqrt{3}}, 0\leq x\leq 3$

$f(x)=\tan^{-1}\left(\frac{x}{3}\right)\Rightarrow$ Range of $f(x)$ is $\left[0,\frac{\pi}{4}\right]$

11. Circle on $\overleftrightarrow{AB}$ diameter is $x^2+y^2-2x-2y=0$

Line $mx-y+2m=0$ will be secant $\Rightarrow d<r$

$\left|\frac{m.1-1+2m}{\sqrt{m^2+1}}\right|<\sqrt{2}$

$9m^2-6m+1<2m^2+2,\ 7m^2-6m-1<0,\ 7m^2-7m+m-1<0$

$7m(m-1)+1(m-1)<0$

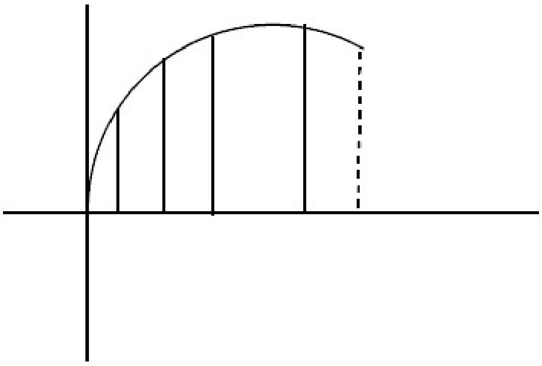
$(m-1)(7m+1)<0$

$m\in\left(-\frac{1}{7},1\right)$

12. Let $f(x)=\sin\left(\frac{\pi x}{2}\right)$

$f(x)$ is increasing in $[0,1]$ and $\int_0^1 \sin\frac{\pi x}{2}dx=\frac{2}{\pi}$

So, $S<\frac{2}{\pi}<R$



13.

$$\text{Volume} = \left|\left[2\vec{b}\times\vec{c}\ \ 3\vec{c}\times\vec{a}\ \ 4\vec{a}\times\vec{b}\right]\right| = 18$$

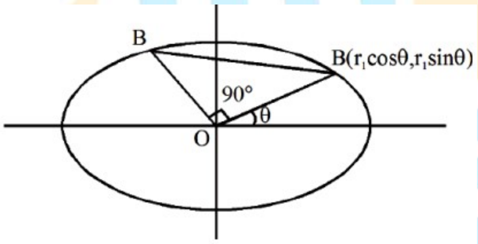
$$24\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = 18 \Rightarrow \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right| = \frac{\sqrt{3}}{2}$$

$$\text{Now, } \left[\vec{a}\vec{b}\vec{c}\right]^2 = \begin{vmatrix} (1+\sin\theta) & \cos\theta & \sin 2\theta \\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right) \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix}$$

Applying $R_1 \to R_1 + R_2 + R_3$ we get after simplification,

$$\left|\left[\vec{a}\vec{b}\vec{c}\right]\right| = \frac{\sqrt{3}}{2}\left|\cos 3\theta\right| = \frac{\sqrt{3}}{2} \Rightarrow \cos 3\theta = \pm 1 \Rightarrow 3\theta = \pi \Rightarrow \theta = \frac{\pi}{3}$$

14.



Let $OA = r_1$ and $OB = r_2$

$\Rightarrow A = (r_1\cos\theta, r_1\sin\theta)$

A lies on ellipse $\frac{x^2}{9} + \frac{y^2}{\lambda^2} = 1$

$$\Rightarrow \frac{r_1^2\cos^2\theta}{9} + \frac{r_1^2\sin^2\theta}{\lambda^2} = 1$$

$$\Rightarrow \frac{1}{r_1^2} = \frac{\cos^2\theta}{9} + \frac{\sin^2\theta}{\lambda^2} \qquad \text{......... (1)}$$

Now replace θ by $90+\theta_1$ we get

$$\Rightarrow \frac{1}{r_2^2} = \frac{\sin^2\theta}{9} + \frac{\cos^2\theta}{\lambda^2} \qquad \text{......... (2)}$$

$$(1) + (2) \Rightarrow \frac{14}{45} = \frac{1}{9} + \frac{1}{\lambda^2}$$

$$\Rightarrow \frac{9}{45} = \frac{1}{\lambda^2} \Rightarrow \lambda^2 = 5$$

15.

$\alpha+\beta+\gamma=0$, $\alpha\beta+\beta\gamma+\gamma\alpha=6$, $\alpha\beta\gamma=-3$

We get $A=\cos^{-1}(\sin(2))=\frac{\pi}{2}-\sin^{-1}(\sin 2)=2-\frac{\pi}{2}$

$B=1$ and $C=\sec^{-1}(\operatorname{cosec}10)=\frac{\pi}{2}-\operatorname{cosec}^{-1}(\operatorname{cosec}10)=10-\frac{5\pi}{2}$

$5A+2B-C=2$

16.

Dividing given expression with $\cot^2\alpha\cot^2\beta\cot^2\gamma$

$2+\tan^2\alpha+\tan^2\beta+\tan^2\gamma=\tan^2\alpha\tan^2\beta\tan^2\gamma$

$2+\operatorname{cosec}^2\alpha-1+\operatorname{cosec}^2\beta-1+\operatorname{cosec}\gamma-1=(\operatorname{cosec}^2\alpha-1)(\operatorname{cosec}^2\beta-1)(\operatorname{cosec}^2\gamma-1)$

$\Rightarrow \sec^2\alpha\sec^2\beta+\sec^2\beta\sec^2\gamma+\sec^2\gamma\sec^2\alpha=\sec^2\alpha\sec^2\beta\sec^2\gamma$

$\Rightarrow \frac{1}{\sec^2\alpha}+\frac{1}{\sec^2\beta}+\frac{1}{\sec^2\gamma}=1$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

17.

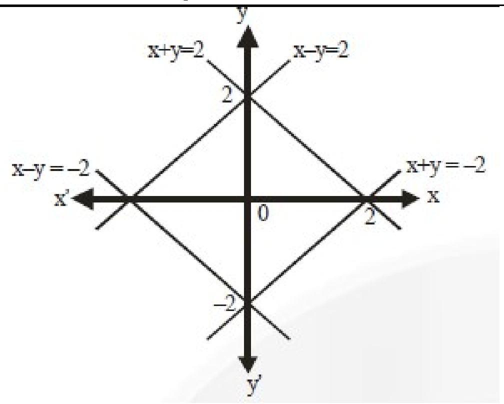
$$\lim_{x\to\infty}\sqrt{x}\left(\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}-\sqrt{x+\sqrt{x}}\right)\Rightarrow \lim_{x\to\infty}\frac{\sqrt{x}\left(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right)}{\sqrt{x}\left(\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}+\sqrt{x+\sqrt{x}}\right)}$$

$$\Rightarrow \lim_{x\to\infty}\frac{\sqrt{x+\sqrt{x}}}{\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}+\sqrt{x+\sqrt{x}}}\frac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}+\sqrt{x}}$$

$$\Rightarrow \lim_{x\to\infty}\frac{\sqrt{1+\frac{1}{\sqrt{x}}}}{\sqrt{1+\sqrt{\frac{x+\sqrt{x\sqrt{x}}}{x^2}}}+\sqrt{1+\frac{1}{\sqrt{x}}}}\frac{1}{\sqrt{1+\sqrt{\frac{x+\sqrt{x}}{x^2}}}+1}=\frac{1}{4}$$

18. $2\ge \max.\{|x-y|,|x+y|\}$

$\Rightarrow |x-y|\le 2$ and $|x+y|\le 2$, which forms a square of diagonal length 4 units.



$\Rightarrow$ The area of the region is $\frac{1}{2}\times 4\times 4=8$ sq. units

This is equal to the area of the square of side length $2\sqrt{2}$

PHYSICS

19. τ_1=RC

$\tau_2=\frac{L}{R}$

$\Rightarrow LC=t_1t_2=0.1\sec$

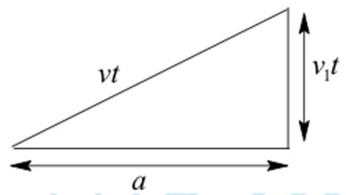
$\Rightarrow T=2\pi\sqrt{LC}=2\pi\sqrt{\frac{1}{10}}=2\sec$

20. $Heat=700\times 10^3\times 1.6\times 10^{-19}\times\frac{dn}{dt}=10\times 10^{-3}$

$\frac{dN}{dt}=\frac{10^{-1}}{10^{-14}}\times\frac{1}{7\times 16}=\frac{10^{12}}{11.2}=\lambda N_0$

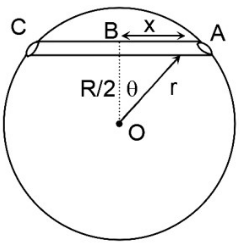
$\lambda=\frac{\ln 2}{14\times 86400}\Rightarrow N_0=\frac{14\times 86400\times 10^{12}}{11.2\ln 2}=154\times 10^{15}$

21. $a^2+v_1^2t^2=v^2t^2$



$t=\frac{a}{\sqrt{v^2-v_1^2}}$

22. $F=-\left(\frac{GM}{R^3}r\frac{X}{r}-\frac{\mu GM}{R^3}\frac{R}{2}\right)$



$$a = -\frac{g}{R}x + \frac{\mu g}{2}$$

$$a = 0 \text{ at } x = \frac{\mu R}{2} = \frac{\sqrt{3}}{2}\frac{R}{2} = \frac{\sqrt{3}R}{4}$$

$$AB = \sqrt{R^2 - \frac{R^2}{4}} = \frac{R\sqrt{3}}{2}$$

$\therefore$ equilibrium position point ‘P’ is midway of AB

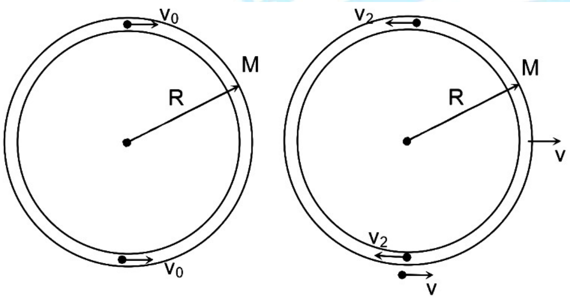
Apply work energy theorem for AP

$$V = \frac{\sqrt{3gR}}{4}$$

23. Conservation of momentum

$$2mv_0 = Mv - 2mv_2$$

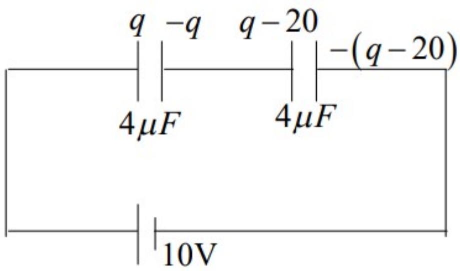
Conservation of energy



$$2\times\frac{1}{2}mv_0^2 = \frac{1}{2}Mv^2 + 2\times\frac{1}{2}mv_2^2$$

$$\Rightarrow v = \frac{4mv_0}{M+2m} = \frac{2v_0}{3}$$

24. $\frac{q}{4} + \frac{1}{2}(q-20) = 10$



$$q + 2q - 40 = 40\mu C$$

$$3q = 80\mu C \Rightarrow q = \frac{80}{3}\mu C$$

$$\text{Charge flow } = \frac{80}{3} - 20 = \frac{20}{3}\mu C$$

$$\omega_{battery} = \frac{20}{3}\times 10 = \frac{200}{3}\mu J$$

25. Electric field at distance x from the center (outside the region) is given by

$$E=\frac{kR^2}{2x}\Rightarrow \text{force required } F=\frac{kqR^2}{2x}$$

Take line integral of dot product of force and displacement along a straight line from point P of infinity.

26. Tension in the string will always be zero.

27. **Conceptual**

28. Field due to ring on Y-axis has to be perpendicular to Y-axis, since net charge on the ring is zero.

29. $\varepsilon=-\frac{d}{dt}\phi$, for charge accumulation at junction applying Gauss law.

30. **Conceptual**

31. **Conceptual**

32. **Conceptual**

33. Apply Lens maker formula for two half lens separately

34. **Conceptual**

35. **Conceptual**

36. nhv=lA

$$n=\frac{lA}{hv}$$

$$l_P=\frac{ne}{10^5}$$

$$1\times10^{-3}=\frac{e}{10^5}\frac{lA}{h\nu}$$

Also, $eV_0=h\nu-\phi$

$$h\nu=\phi+eV_0=(3.5+2.5)eV=6\ eV$$

$$I=\frac{10^{-3}\times10^5\times6}{15\times10^{-4}}=4\times10^5\ \ W/m^2$$

CHEMISTRY

37. $K.E=\frac{1}{2}mv^2$

$$d(KE)=mv.dv$$

$$dv=\frac{d(KE)}{mv}\ (i)$$

$$\Delta x=\frac{h}{4\pi m\Delta v}\ (ii)$$

$$\Delta x=\frac{h}{4\pi m\frac{d(KE)}{mv}}$$

$$=\frac{6.62\times10^{-34}\times2\times10^{6}}{4\pi\times\frac{6.62}{\pi}\times10^{-21}}=500\overset{0}{\mathrm{A}}$$

38. $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}\Rightarrow\frac{20\times15}{T_1}=\frac{10\times60}{T_2}\Rightarrow\frac{T_2}{T_1}=\frac{6}{3}=2$

$$\Delta S=2.303\times n\left[C_P\log\frac{T_2}{T_1}+R\log\frac{P_1}{P_2}\right]$$

$$\Delta S=2.303\times1\left[(30.96\times\log2)+8.314\times\log\frac{20}{10}\right]$$

$$\Delta S=21.3624+5.73666=27.09$$

39. $2NaBH_4+I_2\rightarrow B_2H_6+2NaI+H_2$

$$2BH_3+6NaOH\xrightarrow{450K}B_2H_6+2NaF$$

$$\mathrm{B_2H_6+2CO\longrightarrow2[H_3B\leftarrow CO]}$$

40. $X=XeO_3;Y=XeO_2F_2$

41. Conceptual

42. Arrhenius equation is $k=A\times e^{\left(\frac{-E_a}{RT}\right)}$

On comparing the given equation,

$$\frac{-E_a}{R}=-2700\Rightarrow E_a=(2700\times2)=5400\,cal=5.4\,k.cal$$

43. Conceptual

44. Conceptual

45. A) CO_2 gas liberates

B) SO_2 gas liberates

C) CO_2+CO gases liberates

D) NO_2 gas liberates

46. Conceptual

47. Conceptual

48. Conceptual

49. Conceptual

50. $mpv=\sqrt{\frac{2RT}{M}}\Rightarrow M=\frac{2\times8\times450}{600\times600}=0.02\,Kg/mole=20\,g/mol$

51. From the data, $\mathrm{P_B^o}=30$ and $\mathrm{P_A^o}=40$

$\mathrm{n_A}=\frac{\mathrm{w}}{80};\quad\mathrm{n_B}=\frac{\mathrm{w}}{120}$

$$x_A = \frac{\frac{w}{80}}{\frac{w}{80}+\frac{w}{120}} = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{3}} = \frac{\frac{1}{2}}{\frac{5}{6}}\ ; \quad x_A = \frac{3}{5}$$

$$P_{Total} = 40(3/5)+30(2/5) = 24+12 = 36 = x$$

$$\therefore \frac{X}{6} = \frac{36}{6} = 6$$

52. $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

53. a = 4; b= 2

54. Conceptual